

Fine-Grained Credit Assignment Through Trajectory Branching

Q: What is 6 x 3?

GRPO (No Branching)

Q T $6+6=12$, $12+6=\boxed{18}$ ✓
└ $6+6=42$, $42+6=\boxed{48}$ ⚠ should not blame the last step
└ $6+6=12$, $12+6=\boxed{48}$ ⚠ should not blame the first step
└ $6+6=42$, wait, actually, $6+6=12$, $12+6=\boxed{18}$ ⚠ should not credit the first step

Spontaneous Branching

Q T OK. Let's do it step by step... ⚠ Trajectories can diverge too soon
└ us do it step by step... and produce no meaningful signal
└ Let's first calculate...

Win-Rate or Heuristic-Based Credit Assignment

Q - $6+6=12$. T $12+6=\boxed{18}$ ✓ High contrast for near-leaf correctness divergence
└ $12+6=\boxed{42}$ ✓ High contrast for near-leaf correctness divergence
└ $12+6=18$. T Therefore, the answer is $\boxed{18}$. ⚠ If leaf siblings agree, the contributor is more likely parent than leaves.
└ So, $6 \times 3 = \boxed{18}$.

Our TreeMAPPO Algorithm

Q - $6+6=12$. T $12+6=\boxed{18}$ ✓ High contrast for near-leaf correctness divergence
└ $12+6=\boxed{42}$ ✓ High contrast for near-leaf correctness divergence
└ $12+6=18$. T Therefore, the answer is $\boxed{18}$. ✓ If leaf siblings agree, the contributor is more likely parent than leaves.
└ So, $6 \times 3 = \boxed{18}$.
└ ``python 12+6`` ``18`` So, $6 \times 3 = \boxed{18}$ ✓ Supports token-level branching and tool call handling